

LESSON 7-5 • HOMEWORK PACKET • LEVEL 1 • SHEET 1 OF 2

Plotting Points on the Coordinate Plane

Maryland Standard 6.NS.C.6c · Learning target – I can plot a point with rational coordinates and name the quadrant it lies in.

REMEMBER

- Start at the **origin** (0, 0). Move x units right or left, then y units up or down.
- Quadrant signs: I (+, +), II (–, +), III (–, –), IV (+, –).
- A point with a 0 coordinate is **on an axis**, not in a quadrant.

WORKED EXAMPLE

Plot $(-3.5, 2)$ and name its quadrant.

Step 1 Start at the origin. Move left 3.5 units.

Step 2 Move up 2 units. Left and up is Quadrant II.

Answer Quadrant II

WORD BANK

ordered pair – (x, y) : first move right or left, then up or down

origin – (0, 0), where the axes cross

quadrant – one of the four parts of the grid

on an axis – a point with a 0 in it

PRACTICE

show your work · label units

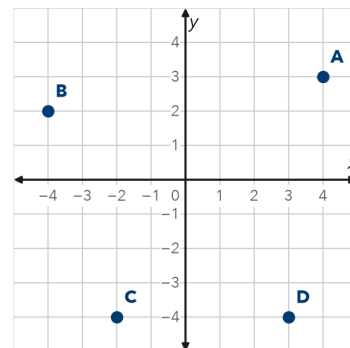
1. Write the ordered pair for each point. Name its quadrant.

Step 1 Start at the origin. Count left or right to the point: that is x .

Step 2 Count up or down: that is y .

Step 3 Use the signs to name the quadrant.

Point	Ordered pair	Quadrant
A		
B		
C		
D		

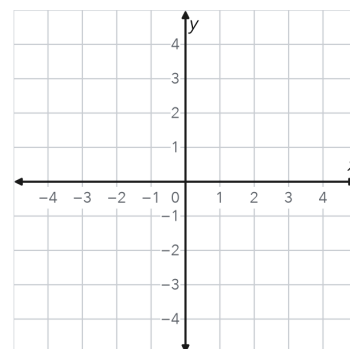


2. **Town Map.** The library is at the origin. Plot and label each place: Park P(2, 4), Pool Q(–4, –1), School S(–3.5, 3), Garden G(4, –2.5).

Step 1 Move right or left first (the first number).

Step 2 Then move up or down (the second number).

Which place is in Quadrant III? _____ Quadrant IV? _____



3. Name the quadrant or axis for each point.

Step 1 Look at the sign of each number.

Step 2 If a number is 0, the point is on an axis.

a. $(-7, -2)$ _____

b. $(8, -1)$ _____

c. $(0, 6)$ _____

d. $(-1.5, 4)$ _____

e. $(-9, 0)$ _____

f. $(2.5, -6)$ _____

4. Reasoning. Malia says $(-3, 5)$ and $(5, -3)$ name the same point. Is she right? Explain.

Frame Malia is ____ because _____. $(-3, 5)$ is in Quadrant ____ and $(5, -3)$ is in Quadrant ____.

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #_____

AT HOME With someone at home, name a quadrant. Take turns saying an ordered pair that lands in it.

LESSON 7-5 • HOMEWORK PACKET • LEVEL 1 • SHEET 2 OF 2**Reflecting Points Across Axes**

Maryland Standard 6.NS.C.6b · Learning target – I can reflect a point across the x-axis or the y-axis and name the new coordinates.

REMEMBER

- **x-axis** reflection: flip the sign of y . $(3, 4)$ becomes $(3, -4)$.
- **y-axis** reflection: flip the sign of x . $(3, 4)$ becomes $(-3, 4)$.
- Across **both** axes, both signs flip.

WORKED EXAMPLE

Reflect $(-2, 5)$ across the x-axis.

Step 1 Keep $x = -2$.

Step 2 Change the sign of y : 5 becomes -5 .

Answer $(-2, -5)$

WORD BANK

reflection – a mirror flip

across the y-axis – change the sign of x

across the x-axis – change the sign of y

PRACTICE

show your work · label units

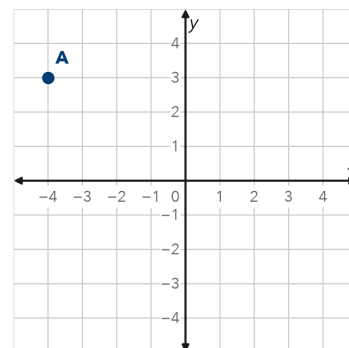
1. Point A is at $(-4, 3)$. Plot point B, the reflection of A across the x-axis. Plot point C, the reflection of A across the y-axis.

Step 1 For B, keep x . Change the sign of y .

Step 2 For C, keep y . Change the sign of x .

a. B = _____

b. C = _____



2. Write the new point after each reflection.

Step 1 Find the mirror: the x-axis or the y-axis.

Step 2 Change the sign of only one number. For both axes, do one flip, then the other.

a. $(-6, -2)$ across the y-axis _____

b. $(-4, 7)$ across both axes _____

3. **Park Map.** A bench is the **mirror image** of a bus stop at $(7, -2.5)$. The mirror is the **x-axis**. Where is the bench?

Answer _____

4. **Stretch.** Point K was reflected across the x-axis. The new point is $(-6, -5)$. Where was point K?

K _____

- 5. Find the Error.** Sam reflected $(-3, -8)$ across the y -axis and got $(-3, 8)$. Explain his mistake. Fix it.

Frame Sam changed the ____-coordinate. Across the y -axis, the ____-coordinate changes sign. The correct point is ____.

TEST PRACTICE

choose the best answer

- 6.** Point $M(-5, 2)$ is reflected across the y -axis. What is the new point?
A $(-5, -2)$ **B** $(5, 2)$ **C** $(5, -2)$ **D** $(2, -5)$
- 7.** The points $(6, -1)$ and $(6, 1)$ are reflections of each other across –
A the x -axis **B** the y -axis **C** both axes **D** neither axis

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #____